Backup and Archival

restore the system, keep the record

Watch the 4-min video

In one line: backups restore a working system quickly after failure; archives keep rarely needed records for years. Design archives with a searchable catalog, an archival tier matched to how fast you'll need data back, WORM retention, copies in another region *and* account, deletion when retention ends, and restore tests.

1 The missing invoice

An auditor asks for an invoice from six years ago. Old rows were deleted from the production database long ago to keep it fast. The backup system keeps 90 days. The document is gone, and the fine is real. The team had backups, but no **archive**: the two sound similar, solve different problems, and are designed differently.

2 Backup vs archive

Backup: restore a working system
copies of live data, taken often (minutes to daily)
kept weeks to months, rotated
measured by RPO / RTO: restore fast
e.g. DB snapshots + PITR, volume snapshots

Archive: keep the record
data moved out of live systems: rarely read
kept years, often by law, then deleted
measured by findability + integrity
e.g. invoices, contracts, audit logs, old media

a 90-day backup is not an archive, and a 7-year archive won't get your database back by lunchtime

Figure 1. Same storage technology, different goals, retention and success measures.

- **RPO** (recovery point objective): how much data you can afford to lose, which sets backup frequency. Continuous WAL/binlog archiving gives point-in-time recovery to within seconds; nightly snapshots give up to 24 hours of loss.
- **RTO** (recovery time objective): how long a restore may take. It decides where backups live: a multi-terabyte restore from a deep archive tier can't meet a 1-hour RTO.
- **3-2-1**: at least 3 copies, on 2 different media or services, 1 off-site. The modern **3-2-1-1-0** adds 1 immutable or offline copy and 0 errors in restore verification.
- **Replication is not a backup**: it faithfully copies deletes and corruption. Backups are point-in-time copies you can go back to.
- Backup retention usually rotates (e.g. 7 daily, 4 weekly, 12 monthly); archive retention is driven by law or policy and is per record.

3 Archival tiers and retrieval times

For Glacier Flexible and Deep Archive, reading is a two-step process: `RestoreObject` creates a **temporary copy** for the number of days you ask for, and you `GET` that copy. Every retrieval is billed per GB and per request, bulk being cheapest. These two classes also add about **40 KB of overhead per object** (8 KB billed at Standard for the name and metadata, 32 KB at the archive rate), so millions of tiny

files should be bundled (e.g. one tar per day) before archiving, and the bundle's contents indexed in the catalog.

Tier	Retrieval	Minimum duration	≈ $/GB-month*
S3 Glacier Instant Retrieval	milliseconds (no restore step)	90 days	0.004
S3 Glacier Flexible Retrieval	expedited 1–5 min · standard 3–5 h · bulk 5–12 h	90 days	0.0036
S3 Glacier Deep Archive	standard within 12 h · bulk within 48 h	180 days	0.00099
Azure Blob Archive	rehydrate: standard up to 15 h · high priority often < 1 h (< 10 GB)	180 days	≈ 0.001
GCS Archive	milliseconds, high retrieval fee	365 days	≈ 0.0012

*Approximate list prices in a major US region; retrieval and request fees come on top. Check current pricing.

4 Match the tier to how you'll retrieve

Choose by the retrieval promise, not by importance. Records an auditor may want within a day can sit in Glacier Flexible (standard restore in hours) or Deep Archive (within 12 h). Records you might need in seconds (customer self-service downloads of old statements) belong in Glacier Instant Retrieval or a cool tier. Records you'll almost certainly never read but must keep go to the deepest tier. And always build a small, searchable **catalog** (a database table) of what's archived: record ID, business keys (customer, date, type), object key, checksum, retention date. Without it, finding six invoices means restoring and opening millions of objects. The catalog itself needs backups.

Store archives in open, long-lived formats (PDF/A, CSV, Parquet, JSON) with their schema, so they're readable in ten years without today's software.

5 Cross-region, cross-account copies

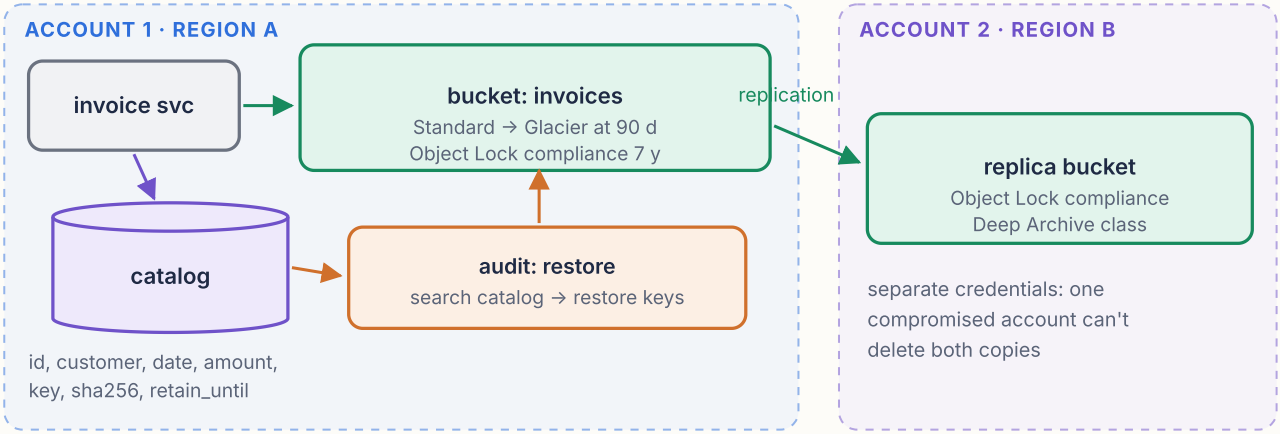


Figure 2. The 7-year archive: locked primary copy plus a locked replica in another region and another account, and a catalog to find records.

One region is one failure domain; one account is one blast radius (a leaked admin key, a malicious insider, a bad automation run). S3 **Cross-Region Replication** copies new objects asynchronously to a bucket in another region; both buckets need versioning, existing objects need S3 Batch Replication, **Replication**

Time Control adds an SLA of 99.99% of objects within 15 minutes, and the replica can land directly in a Glacier class. Replicating into **another account** with ownership transferred to it means credentials for account 1 can't delete the replica. By default S3 replication does not replicate deletions of specific versions, so a malicious version delete in the source doesn't propagate. Azure offers geo-redundant storage (GRS/RA-GRS) and object replication; AWS Backup and Azure Backup can copy backups across regions and accounts.

6 Tamper-proof, and tested

- **WORM**: S3 Object Lock in **compliance** mode (nobody, including root, can shorten retention or delete a locked version), Azure immutable storage with a locked time-based policy, AWS Backup Vault Lock, Azure Backup immutable vaults. This protects records from mistakes, insiders and ransomware, which now routinely tries to delete backups first. Enable it on the replica too.
- **Integrity**: store checksums in the catalog and verify them on restore; periodically restore a random sample and re-hash it.
- **Restore tests**: a backup nobody has restored is only a hope. Schedule restore drills (AWS Backup has a restore-testing feature), measure the real RTO, and test the runbook, permissions and people, not just the bytes.
- **Deletion**: when retention ends, a lifecycle rule deletes the record (lesson 12), unless a legal hold says otherwise. Keeping data longer than required is a liability and a cost.

7 Example: a 7-year archive of financial documents

WORKED EXAMPLE: INVOICES KEPT FOR 7 YEARS

Each invoice PDF (~200 KB) is written to `s3://invoices/2026/11/13/INV-88412.pdf` with Object Lock **compliance** retention until creation + 7 years, and a catalog row (customer, date, amount, key, SHA-256, retain_until). After 90 days a lifecycle rule moves it to Glacier Flexible Retrieval; replication sends a copy to region B, account 2, stored in Deep Archive with the same lock. At 1 million invoices a year, the archive reaches 7 million objects and ~1.4 TB: roughly $5 a month in Glacier Flexible, plus about 280 GB of per-object overhead (7 M × 40 KB), a 20% surcharge worth knowing about. At this size the tier hardly matters in dollars; the retrieval promise decides.

An auditor asks for ACME's 2020 invoices: the catalog returns 42 keys, the API issues 42 standard restores, and 3–5 hours later the PDFs are delivered with verified checksums, without restoring anything else. After 7 years the lock expires and a lifecycle rule deletes the record from both copies.

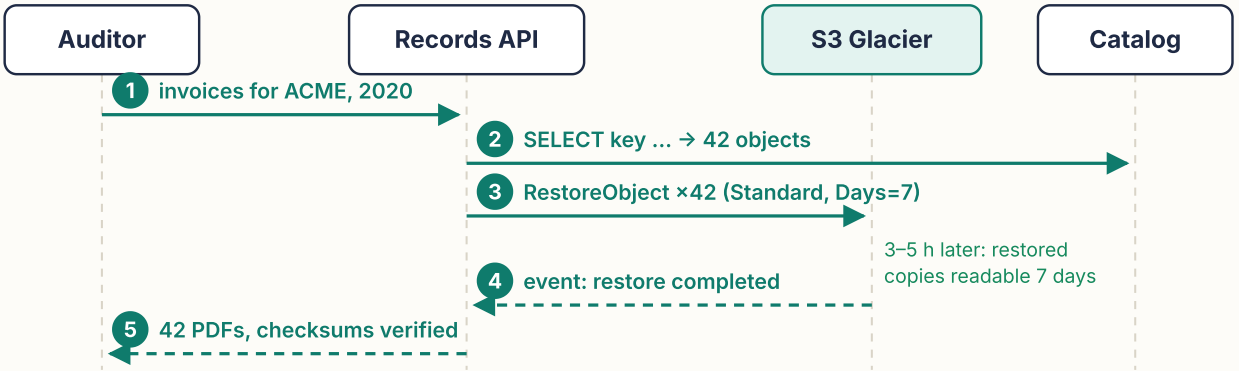


Figure 3. An audit request: query the catalog, restore only the matching objects, verify checksums, deliver.

TRADE-OFFS

- Deeper tiers cost less to keep and more, and longer, to read; minimum durations penalise early deletes.
- Compliance-mode WORM is the strongest protection and is irreversible: a wrong retention period is paid for years.
- Cross-account replication limits blast radius but doubles storage and adds key and permission management.
- Bundling small files cuts per-object overhead but makes single-record retrieval coarser.

COMMON MISTAKES

- Treating backups as the archive (90-day rotation deletes the records you must keep).
- Archiving without a catalog, so finding one record means restoring everything.
- Keeping every copy in one account, where one compromised credential can delete them all.
- Never testing a restore, then discovering missing permissions or a 2-day retrieval during an incident.

8 Key terms

RPO / RTO: acceptable data loss / acceptable restore time

Restore / rehydrate: making an archived object readable again, for a fee

Cross-region replication: asynchronous copy of objects to another region

Archive tier: cheap, slow-to-read storage (Glacier, Azure Archive)

3-2-1(-1-0): 3 copies, 2 media, 1 off-site (+1 immutable, 0 errors)

Compliance-mode Object Lock: WORM retention no one can shorten or remove

9 Interview questions

Q1. What's the difference between a backup and an archive? BASIC

A backup restores a working system after a failure, so it's frequent, short-lived and fast to restore; an archive keeps rarely used records for the long term, findable and intact.

Q2. What are RPO and RTO? BASIC

RPO is how much recent data you can lose (sets backup frequency); RTO is how long recovery may take (sets where and how backups are stored).

Q3. Design an archival strategy for compliance documents. SCENARIO

Object storage plus a searchable catalog, lifecycle moves to an archival tier by age, WORM retention (Object Lock compliance mode), replication to another region and account, restore tests, and deletion when retention ends.

Q4. Ransomware encrypted production and tried to delete backups. What design makes recovery possible? SCENARIO

Immutable backups (Vault Lock / Object Lock / immutable vaults) in a separate account with separate credentials, plus regularly tested restores within the RTO.

Q5. You must archive 500 million small log files. What changes? SCENARIO

Bundle them into large compressed files before archiving to avoid per-object overhead and request costs, and index file-to-bundle locations in a catalog.

Q6. Why isn't cross-region replication alone a backup? TRICKY

It replicates deletes and corruption too, and has no history to go back to; you need versioning, point-in-time copies and immutability.

Q7. A team wants their 5 TB database backups in Deep Archive to save money. Problem? TRICKY

Restores take up to 12–48 hours plus retrieval costs, which breaks any normal RTO; keep recent backups in a fast tier and only age older ones down.

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